

Matthew J. Cockayne

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RESEARCH INTERESTS

Safe and responsible AI, ensuring AI systems are interpretable, robust, and equitable. My research focus is developing methods for bias detection and mitigation, systematic robustness evaluation under distribution shift, and interpretable-by-design architectures that enable human oversight and intervention. Applied through medical imaging, where deployment failures carry direct real-world harm, I produce domain-agnostic frameworks transferable to broader safe and responsible AI challenges.

EDUCATION

PhD in Computer Science 2023-2026 (Expected)

Keele University, Keele, UK

Thesis: Responsible AI for Healthcare: Robustness, Interpretability, and Fairness in Skin Lesion Recognition

Funding: Competition-funded Department Studentship

Published 2 first-author papers (Pattern Analysis and Applications journal, AIIH 2025 conference oral presentation). Two additional first-author papers submitted to MICCAI 2026 and Image and Vision Computing (Elsevier). Co-author on Discover Computing journal paper under review. Developed open-source toolkits MedXAI and SkinCBM for reproducible research. PGR Representative for 3 consecutive years.

MSc in Artificial Intelligence and Data Science, Distinction 2021-2022

Keele University, Keele, UK

Data Analyst Internship: Developed explainable AI methods for CV-based sentiment analysis in recruitment.

Turing Network Development Research Associate: Organised networking events and hackathons.

BSc in Physics, First Class Honours 2018-2021

Keele University, Keele, UK

RESEARCH EXPERIENCE

PhD Researcher - Responsible & Trustworthy AI 2023-Present

Keele University, School of Computer Science and Mathematics

- Designed robustness evaluation frameworks for transformer-based architectures, systematically benchmarking model behaviour under 15+ real-world corruption types (noise, blur, compression) and quantifying failure modes through uncertainty estimation; applied to multi-modal skin cancer detection achieving state-of-the-art performance while identifying deployment risks

Code: github.com/xrai keele/DermFormer

- Developed interpretable-by-design architectures (concept bottleneck models) enabling human-in-the-loop oversight. Users can inspect, validate, and correct intermediate reasoning steps, ensuring AI decisions remain auditable and aligned with human intent

- Conducted information-theoretic analysis of model completeness and concept synergy, revealing when models rely on spurious feature correlations versus genuine causal reasoning; implemented sparse polynomial classifiers preserving full interpretability of decision boundaries

- Built fairness-aware curriculum learning reducing demographic performance disparities by 44% and improving worst-group outcomes by 63%, demonstrating that safety-oriented training objectives can improve rather than constrain overall system performance

- Evaluated foundation model (SAM) reliability in zero-shot transfer settings, characterising failure modes and developing adaptive selection methods reducing catastrophic failures by 98.75%

Code: github.com/xrai keele/Classification-to-Segmentation

Data Analyst Intern - Explainable AI 2021-2022

Industry Partner via Keele University

- Developed explainable AI techniques for NLP models analysing sentiment in recruitment contexts

- Implemented interpretability methods ensuring transparency and fairness in automated decision-making systems

PUBLICATIONS

Journal Articles

Cockayne, M. J., Ortolani, M., Al-Bander, B. (2025). “DermFormer: Nested Multi-modal Vision Transformers for Robust Skin Cancer Detection.” *Pattern Analysis and Applications*, Springer Nature, Volume 28, Article 194.

DOI: [10.1007/s10044-025-01572-0](https://doi.org/10.1007/s10044-025-01572-0)

Cockayne, M. J., Ortolani, M., Al-Bander, B. “Sparse Concept Absence Network for Concept Bottleneck

Models.” *Image and Vision Computing*, Elsevier. Status: Submitted.

Al-Bander, B., Ortolani, M., Dafaalla, M., **Cockayne, M. J.**, Pergola, G., Mamas, M. A. “Investigating and Mitigating Bias in Cardiovascular Disease Mortality Predictors for Scaling-up AI Fairness.” *Discover Computing*, Springer Nature. Status: Under Review.

Guo, S., **Cockayne, M. J.**, Ortolani, M., Al-Hamd, R., Al-Bander, B. “Zero-Shot Crack Segmentation for Structural Health Monitoring Using Class Activation Map Guided Segment Anything Model.” *Machine Vision and Applications*. Status: Under Review.

Conference Papers

Cockayne, M. J., Ortolani, M., Al-Bander, B. (2025). “Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation.” *AIHH 2025*, University of Cambridge. [Oral Presentation] DOI: [10.1007/978-3-032-00656-1.24](https://doi.org/10.1007/978-3-032-00656-1.24)

Cockayne, M. J., Ortolani, M., Al-Bander, B. “Fair Curriculum Learning for Concept Bottleneck Models in Dermatology.” *MICCAI 2026*. Status: Submitted.

TECHNICAL SKILLS

Programming: Python, LaTeX, R, SQL

Safe & Responsible AI: Fairlearn, Aequitas, adversarial debiasing, Fairness benchmarking, robustness benchmarking, uncertainty quantification

Explainable AI: CAM, GradCAM, integrated gradients, SHAP, LIME, attention visualisation, concept bottleneck models

ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face

Foundation Models: Segment Anything Model (SAM/MedSAM), zero-shot transfer evaluation, failure mode analysis

Computer Vision: OpenCV, torchvision, TIMM

Data Analysis: Pandas, NumPy, SciPy, statistical modelling, information-theoretic analysis

Development: Git/GitHub, Jupyter Notebooks, Weights & Biases, TensorBoard

High Performance Computing: HPC cluster management, SLURM job scheduling, batch script optimisation

TEACHING EXPERIENCE

Laboratory Demonstrator

2022-Present

Keele University, School of Computer Science and Mathematics

- Supported undergraduate and master’s students across computer science modules for 4 years
- Delivered guest lectures on deep learning, computer vision, and responsible AI
- Mentored students on machine learning and AI ethics research projects

PROFESSIONAL SERVICE

Programme Committee: International Conference on AI in Healthcare (AIiH) 2026

Peer Review: Reviewer, 1st International Workshop on AI Safety and Security (AI-SS) 2026; *Engineering Applications of Artificial Intelligence* journal (2024-present)

Leadership: Postgraduate Research Representative, Keele University (2022-2025, 3 years); Student Representative, Physics Programme (2018-2021, 3 years)

Mentorship: Supervised undergraduate and master’s research projects on machine learning and AI ethics; provided guidance on responsible AI methodologies and fairness evaluation techniques

Outreach: Data Science Content Creator Internship (2021-2022) - developed educational resources and materials on AI and data science fundamentals for broader audiences

Community: Turing Network Development Research Associate (2021-2022) - organised networking events, hackathons, and interdisciplinary research initiatives to foster cross-disciplinary collaboration

PRESENTATIONS AND TALKS

Oral Presentation: “Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation.” AIHH 2025, University of Cambridge (Sep 2025)

Research Seminars: Presented research on safe and responsible AI, robustness evaluation, fairness-aware learning, and interpretable architectures at departmental research seminars, Keele University (2022-present)

Guest Lectures: Deep learning, safe and responsible AI (fairness, interpretability, bias mitigation), robustness evaluation, transformers in high-stakes domains (2022-present)

REFERENCES

Available upon request